

Original Research Paper

A Novel Model to Predict Progression to Death After Withdrawal of Care in Potential Donation-After-Circulatory-Death Heart Donors

AUSTIN AYER, MD¹ PRANEET MYLAVARAPU, MD¹ DAVID GOLOMBECK, MD²
 BIJAL RAJPUT, MD² FADI AZAR, MD¹ BERNARD S. KADOSH, MD² QUAN M. BUI, MD¹
 MARK J. KEARNS, MD³ VICTOR PRETORIUS, MBChB³ MARCUS A. UREY, MD¹
 ANTOINETTE S. BIRS, MD¹ and ERIC D. ADLER, MD¹

La Jolla, California; and New York, New York

ABSTRACT

Background: Predicting successful heart donation after circulatory death (DCD) remains a challenge. We developed a model to predict progression to circulatory death after withdrawal of care in potential DCD heart donors.

Methods and Results: Adult DCD heart offers at a single institution over a 6-month period were retrospectively reviewed. Univariate logistic regression was used to assess associations between pre-withdrawal-of-care variables and a binary outcome of circulatory death. The discriminatory performance of multivariate logistic regression and 4 supervised machine-learning models in predicting circulatory death was assessed by stratified cross validation. The top-performing model type was developed using the full dataset and externally validated in a cohort of potential donors at a distant center. Of 234 included offers, 150 (64.1%) progressed to circulatory death. Factors associated with circulatory death included Glasgow Coma Scale, brainstem reflexes; mean arterial pressure, vasoactive inotropic score, ventilator triggering, positive end-expiratory pressure, partial pressure of arterial oxygen/fraction of inspired oxygen ratio, and serum sodium. A multivariate logistic regression model demonstrated sensitivity of 0.85, specificity of 0.52, and area under the receiver operating characteristic curve of 0.77 in the development cohort, and sensitivity of 1.00, specificity of 0.50, and area under the receiver operating characteristic curve of 0.88 in external validation.

Conclusions: We present a novel model to predict progression to circulatory death among potential DCD heart donors. (*J Cardiac Fail* 2026;00:1–12)

Key Words: Heart transplantation, donation after circulatory death, prediction modeling, donor management.

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Introduction

Donation after circulatory death (DCD) accounts for a growing and significant proportion of transplanted hearts.¹ Numerous studies have shown outcomes after DCD and

donation after brain death heart transplantation to be comparable,^{2–7} and broad adoption of DCD could significantly expand the US heart donor pool and reduce waitlist times.^{8–10} However, DCD presents unique challenges in donor selection and allocation of procurement resources.

From the ¹Division of Cardiovascular Medicine, University of California San Diego, La Jolla, California; ²Department of Medicine, New York University School of Medicine, New York, New York and ³Division of Cardiovascular and Thoracic Surgery, University of California San Diego, La Jolla, California.

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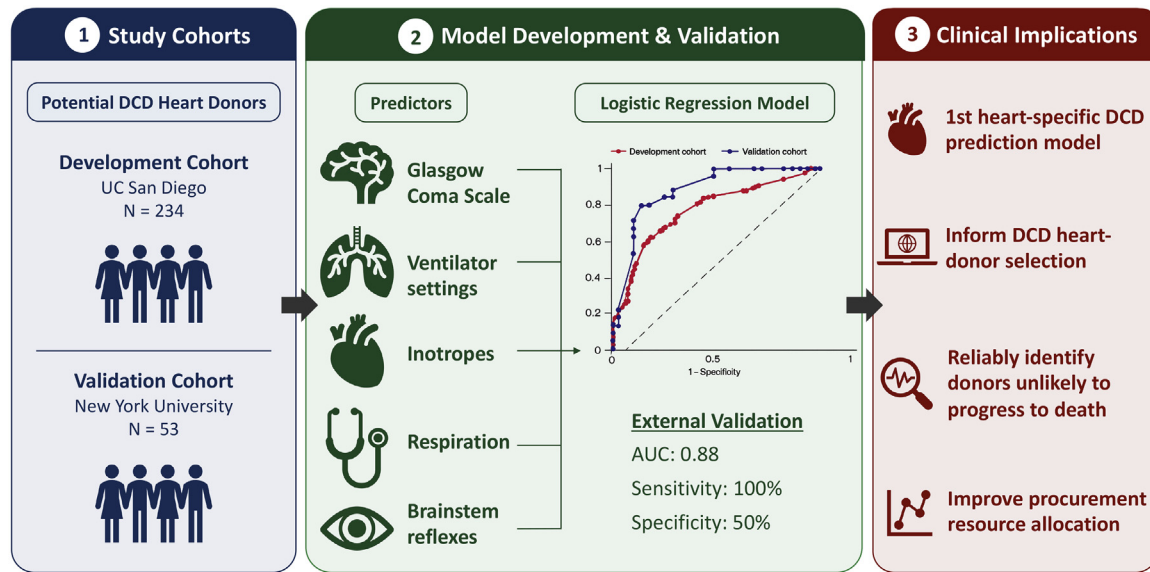
Reprint requests: Antoinette Birs, MD, Sulpizio Cardiovascular Center, 9452 Medical Center Dr, Mail Code 7411, La Jolla, CA 92037. Tel: 858-822-5062. E-mail: abirs@health.ucsd.edu

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Central Illustration

In contrast to donation after brain death, DCD requires withdrawal of donor cardiopulmonary support and subsequent circulatory cessation before organs may be procured.¹¹ The hemodynamic compromise inherent to DCD results in ischemic insult that may affect organ use and transplant outcomes.¹²⁻¹⁴ In addition, not all potential DCD donors progress to death within a time frame conducive to heart donation.¹⁵ Donors may remain hemodynamically stable after withdrawal of care, precluding a feasible procurement timeline. Alternatively, donors may experience a prolonged agonal phase of warm ischemia, resulting in prohibitive organ damage. Given this potential for unsuccessful donation, DCD donor selection remains a challenge.

Multiple clinical tools exist to stratify the likelihood of timely progression from withdrawal of care to death in the general DCD donor population,¹⁶⁻²⁴ but none were developed or validated exclusively in potential DCD heart donors, who comprise a small subset of total DCD donors and must meet different clinical criteria than donors at large.^{11,25}

Unsuccessful DCD procurement incurs significant costs in terms of resource allocation¹ and affects the quality of life for donor families, patients who are waitlisted, and surgical staff. Thus, a better understanding of the likelihood of, and factors associated with, heart-donor progression to circulatory death may inform the efficient expansion of DCD heart transplantation.

Methods

Study Population

Consecutive DCD heart offers made to adult patients at the University of California San Diego between January 1 and June 30, 2023, were retrospectively reviewed. Donor

demographic and clinical data were obtained from the United Network for Organ Sharing (UNOS). Unless otherwise specified, variables represent the final datapoint recorded before withdrawal of care. This research was determined to be not human subjects research by the University of California San Diego institutional review board (no. 810593, determination May 13, 2024).

Statistical Analysis

Baseline donor data were compared descriptively using either χ^2 or Fisher exact test for categorical variables, or Wilcoxon rank-sum or Student t test for continuous variables. Regression imputation was used to address missing data. To identify associations between pre-withdrawal-of-care clinical variables and a binary outcome variable of progression to circulatory death after withdrawal of care, univariate logistic regression was used to calculate odds ratios (OR) with corresponding 95% confidence intervals (95% CIs) with respect to the outcome of progression to circulatory death. *P* values for univariate logistic regression were determined by likelihood ratio test.

Descriptive statistics and univariate analysis were performed using GraphPad Prism (GraphPad Software, Boston, MA). Data are presented as frequency with percentage for nonmissing categorical variables and as mean with standard deviation (SD) or median with 25th and 75th percentiles (Q1, Q3) for continuous variables that were or were not normally distributed, respectively.

Model Development

Multivariate logistic regression was performed to predict progression to circulatory death using backward elimination of candidate model features identified as significantly associated with the outcome of interest on univariate logistic regression. In addition, 4 supervised machine-

learning models were developed to predict progression to circulatory death, with candidate model features being the same as in the multivariate logistic regression. The machine-learning models developed were random forest, in which a random forest classifier was trained to predict the outcome; decision tree, in which a decision tree classifier was trained to predict the outcome; support vector machine, in which a support vector machine classifier with a radial basis function kernel was trained to predict the outcome; and naive Bayes, in which a Gaussian naive Bayes classifier was trained to predict the outcome.

The performance of each model type was evaluated using 5-fold stratified cross-validation to ensure robust performance assessment and minimize bias. Specifically, the dataset was split into 5 equal parts, and each model type was trained on 4 parts and validated on the remaining part, with this process repeated for all 5 possible iterations. Model performance was evaluated using area under the receiver operating characteristic (ROC) curve (AUC), sensitivity, and specificity, with a probability threshold of 0.5 used in all models. The average AUC, sensitivity, and specificity across the 5 validation replicates were calculated and reported for each model. ROC curves were generated and the average ROC curve for each model was plotted. The top-performing model type was then developed on the full development dataset to produce a deployable model, the performance of which was assessed across a range of probability thresholds. The 95% CI for AUC was estimated using the DeLong non-parametric method. Model calibration was assessed using the calibration intercept and slope, and performance was summarized using the Brier score, or the mean squared difference between predicted probabilities and observed outcomes.

Model performance assessment was carried out using Python (Python Software Foundation, Wilmington, DE). Final multivariate logistic regression model development was performed in SPSS (IBM Corp, Armonk, NY). Model calibration was assessed using R, version 4.5.2 (R Foundation for Statistical Computing, Vienna, Austria).

External Model Validation

UNOS data were retrospectively reviewed for consecutive DCD heart offers made to adult patients at New York University between April 2023 and October 2025. The start of this period coincided with the transition to the center's current transplant operations provider, which maintains organ-offer records. Offers for which any model input variable was unavailable on UNOS were excluded from initial analysis. A sensitivity analysis was performed using an expanded validation cohort including offers with missing data (maximum 1 missing model input), applying multiple imputation ($n=5$) to missing variables. Model-predicted probabilities of circulatory death were compared with observed

outcomes, and model performance characteristics are reported. Imputed datasets were pooled using Rubin's rules to obtain final pooled point estimates and 95% CIs for all metrics.

Relevant Definitions

The vasoactive inotropic score (VIS), a metric of total vasoactive medication support, was calculated as dopamine ($\mu\text{g} \cdot \text{kg}^{-1} \cdot \text{min}^{-1}$) + dobutamine ($\mu\text{g} \cdot \text{kg}^{-1} \cdot \text{min}^{-1}$) + 100 $\times$ epinephrine ($\mu\text{g} \cdot \text{kg}^{-1} \cdot \text{min}^{-1}$) + 100 $\times$ norepinephrine ($\mu\text{g} \cdot \text{kg}^{-1} \cdot \text{min}^{-1}$) + 10 $\times$ milrinone ($\mu\text{g} \cdot \text{kg}^{-1} \cdot \text{min}^{-1}$) + 10,000 $\times$ vasopressin (units $\cdot \text{kg}^{-1} \cdot \text{min}^{-1}$) + 50 $\times$ levosimendan ($\mu\text{g} \cdot \text{kg}^{-1} \cdot \text{min}^{-1}$).²⁶

The functional warm ischemic time, which quantifies the period of organ hypoperfusion after withdrawal of care, was defined according to the DCD Heart Trial as the time in minutes from arterial oxygen saturation $<70\%$ or systolic blood pressure <50 mm Hg until cold cardioplegic flush (if recovery via direct procurement) or institution of cardiopulmonary bypass (if recovery via normothermic regional perfusion).⁶

Data-Availability Statement

Because of the sensitive nature of organ-donor data, not all data used in this study are available. A subset of deidentified data is available from the corresponding author upon reasonable request. An online version of our model is available for experimental use at <https://dcdheartprediction.com/>.

Results

Donor Outcomes and Organ Use

A total of 242 qualifying DCD heart offers occurred during the study period. Eight offers were excluded as the result of donor progression to brain death or unavailable UNOS data. Of the remaining 234 donors, 150 (64.1%) progressed to circulatory death whereas 84 (35.9%) did not. Among donors who did not progress to circulatory death, procurement was generally abandoned 120 minutes after withdrawal of care, although occasionally this time period was shortened at the discretion of the procurement team. Among heart offers that progressed to circulatory death, 30 (20.0%) were accepted at the University of California San Diego, 35 (23.3%) were accepted at an outside institution, and 85 (56.7%) were not accepted. Among the 30 accepted offers at the University of California San Diego, 22 hearts (73.3%) were transplanted and 8 hearts (26.7%) were subsequently declined due to organ quality (Fig. 1). The median functional warm ischemic time among donors that progressed to circulatory death was 24.5 (Q1 19, Q3 36) minutes.

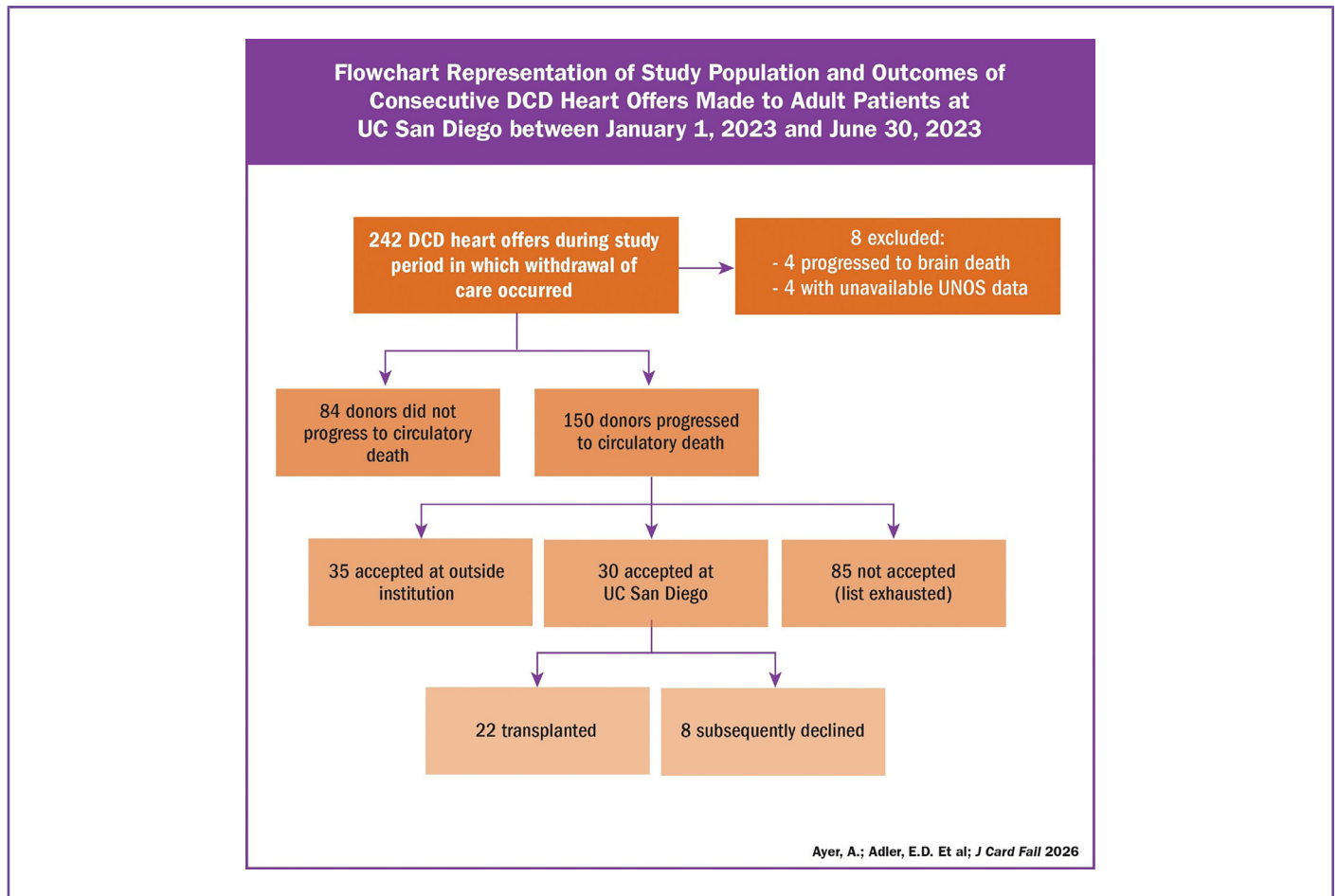


Fig. 1. Flowchart representation of study population and subsequent outcomes of consecutive DCD heart offers made to adult patients at UC San Diego between January 1, 2023, and June 30, 2023. DCD, donation after circulatory death; UNOS, United Network for Organ Sharing.

Study Population and Univariate Analysis

Baseline donor demographic data are displayed in [Table 1](#). Among the model development cohort, mean (SD) age was 37.4 (± 9.7) years, 68.4% were male, and 67.9% were White. Between potential donors who progressed to circulatory death and those who did not, no differences were observed in age, sex, race, body mass index, or cause of neurologic injury.

Factors associated with progression to circulatory death on univariate logistic regression included Glasgow Coma Scale (GCS; OR 0.55, 95% CI 0.41–0.72); absence of corneal (OR 4.09, 95% CI 2.13–8.01), cough (OR 4.50, 95% CI 2.19–10.07), gag (OR 3.61, 95% CI 1.96–6.76), pupillary (OR 3.64, 95% CI 1.89–7.33), or withdrawal-to-pain (OR 3.17, 95% CI 1.66–6.23) reflexes; mean arterial pressure (mm Hg; OR 0.98, 95% CI 0.96 to <1.00); VIS (OR 1.22, 95% CI 1.07–1.48); triggering the ventilator (OR 0.22, 95% CI 0.08–0.51); positive end-expiratory pressure (cmH₂O; OR 1.22, 95% CI 1.07–1.40); partial pressure of arterial oxygen/fraction of inspired oxygen ratio (OR 0.99, 95% CI 0.99–<1.00); and serum sodium (mmol/L; OR 1.04, 95% CI 1.01–1.08) ([Table 2](#), [Fig. 2](#)). On individual

analysis of component medications of VIS, norepinephrine infusion rate ($\mu\text{g} \cdot \text{kg}^{-1} \cdot \text{min}^{-1}$) was positively associated with progression to circulatory death (OR 1.47, 95% CI 1.12–2.31), whereas the vasopressin infusion rate (U/h) was not (OR 1.54, 95% CI 0.92–2.97).

Hospital length of stay, respiratory rate, core body temperature, and serum creatinine were associated with progression to circulatory death by descriptive analysis but not by univariate logistic regression. No association was observed between progression to circulatory death and brain imaging findings or pre-withdrawal-of-care administration of sedative or analgesic medications (assessed as a binary variable representing ongoing administration of a given medication at any dose immediately before withdrawal of care). Among all 39 variables collected, 5.45% of the total discrete data points were missing, and these were missing at random (Little's missing completely at random, $P = .11$).

Model Selection and Performance

In 5-fold stratified cross-validation, multivariate logistic regression demonstrated an average sensitivity of 0.83,

Table 1 Baseline characteristics of potential DCD heart donors who progressed to circulatory death after withdrawal of care compared with those who did not (procurement abandoned)

Variable*	Circulatory Death (n = 150)	Procurement Abandoned (n = 84)	Total (N = 234)	P Value
Age, y	36.8 (10.5)	38.6 (8.1)	37.4 (9.7)	.19
Sex				.31
Female	44 (29.3%)	30 (35.7%)	74 (31.6%)	
Male	106 (70.7%)	54 (64.3%)	160 (68.4%)	
Body mass index, kg/m ²	27.4 (23.4, 32.6)	26.7 (23.5, 32.1)	27.1 (23.4, 32.2)	.77
Race				.81
White	99 (66%)	60 (71%)	159 (68%)	
Hispanic/Latino	27 (18%)	13 (15%)	40 (17%)	
Black/African-American	15 (10%)	8 (10%)	23 (10%)	
Asian	6 (4%)	1 (1%)	7 (3%)	
Indigenous American	3 (2%)	2 (2%)	5 (2%)	
Cause of neurologic damage				.80
Anoxia	86 (57.3%)	50 (59.5%)	136 (58.1%)	
Head trauma	35 (23.3%)	22 (26.2%)	57 (24.4%)	
Cerebrovascular accident	25 (16.7%)	10 (11.9%)	35 (15.0%)	
Other	4 (2.7%)	2 (2.4%)	6 (2.6%)	

DCD, donation after circulatory death

*Categorical data are presented as frequency (%) and continuous data as mean (standard deviation) or median (Q1, Q3).

specificity of 0.54, and AUC of 0.76 with respect to the outcome of circulatory death. Assuming the outcome prevalence observed in the development cohort, multivariate logistic regression achieved average positive and negative predictive values of 76.1% and 63.4%, respectively. Among machine-learning models, the Naïve Bayes model demonstrated the best discriminatory performance, with an average sensitivity of 0.59, specificity of 0.79, and AUC of 0.76, yielding average positive and negative predictive values of 83.0% and 51.6%, respectively. The support vector machine, random forest, and decision tree models demonstrated inferior discriminatory ability, with average AUC of 0.74, 0.71, and 0.63, respectively. Performance characteristics of all candidate models are shown in [Table 3](#) and corresponding average ROC curves are displayed in [Figure 3](#).

A multivariate logistic regression model developed using the complete development cohort and incorporating VIS >5, triggering the ventilator, presence of the pupillary reflex, GCS, and positive end-expiratory pressure demonstrated an AUC of 0.77 (95% CI 0.71–0.83) with respect to the outcome of circulatory death ([Table 4](#)). The model showed significantly improved fit over the intercept-only model (likelihood ratio $\chi^2 = 54.98$, $df = 5$, $P < .001$), with pseudo- R^2 values (Cox & Snell = 0.21; Nagelkerke = 0.29) suggesting moderate explanatory performance. Model performance across a range of probability thresholds is reported in [Table 4](#). Although Youden's index (sensitivity + specificity – 1) was maximized with a probability threshold of 0.7 (sensitivity 0.60, specificity 0.83), this came at the expense of increased false-negative predictions, and a probability threshold of 0.5 (sensitivity 0.85, specificity 0.52) was selected for external

validation due to the significant potential clinical costs (eg, waitlist mortality, unused organs) of false-negative prediction. In the development cohort, apparent calibration was close to optimal, with a calibration intercept of –0.03 (95% CI –0.36 to 0.30) and a slope of 0.99 (95% CI 0.68–1.38). The Brier score was 0.18, representing improvement over the null model based on outcome prevalence (null Brier 0.23).

External Model Validation

Of 110 DCD heart offers assessed for external validation, 53 (48.2%) had all model input variables available and were included in the complete analysis. Within this validation cohort, mean (SD) age was 34.8 (± 8.5) years, 71.7% were male, and 73.6% were White. Donor demographic and clinical data for the validation cohort are available in [Supplemental Table S1](#).

Of 53 included offers, 25 (47.2%) progressed to circulatory death. The multivariate logistic regression model demonstrated a sensitivity of 1.00, specificity of 0.50, and AUC of 0.88 (95% CI 0.78–0.97) with respect to the outcome of circulatory death ([Fig. 4](#)). Assuming the outcome prevalence observed in the external validation cohort, this yielded a positive predictive value of 64% and a negative predictive value of 100%. The calibration intercept was –1.55 (95% CI –2.71 to –0.64) and the calibration slope was 1.51 (95% CI 0.80–2.45), suggesting systematic over-prediction and reasonable distribution of predictions, respectively ([Fig. 5](#)). The Brier score was 0.19 (null Brier 0.25), indicating moderate improvement over the null model.

In sensitivity analysis, the model was applied to an expanded validation cohort of 86 potential donors with

Table 2 Association of pre-withdrawal-of-care variables with progression to circulatory death among potential DCD heart donors as assessed by descriptive analysis and univariate logistic regression

Variable	Circulatory Death (n = 150)	Procurement Abandoned (n = 84)	P Value (Descriptive)	Odds Ratio (95% CI)	P Value (Log. Regression)
Admission duration, d	9 (6, 11)	10 (7, 14)	<.01	0.98 (0.94–1.01)	.23
Previous cardiac arrest	85 (58.2%)	41 (51.9%)	.36	1.29 (0.74–2.24)	.36
Glasgow Coma Scale (range 3–15)	3.0 (3.0, 4.0)	4.0 (3.0, 5.25)	<.0001	0.55 (0.41–0.72)	<.0001
Absence of reflex					
Corneal	78 (72.9%)	25 (39.7%)	<.0001	4.09 (2.13–8.01)	<.0001
Cough	54 (42.9%)	10 (14.3%)	<.0001	4.50 (2.19–10.07)	<.0001
Gag	84 (67.2%)	25 (36.3%)	<.0001	3.61 (1.96–6.76)	<.0001
Pupillary	63 (51.2%)	15 (22.4%)	<.001	3.64 (1.89–7.33)	<.001
Withdrawal to pain	63 (57.8%)	19 (30.2%)	<.001	3.17 (1.66–6.23)	<.001
Triggering ventilator	81 (69.8%)	64 (91.4%)	<.001	0.22 (0.08–0.51)	<.01
Heart rate, beats/min	87.3 (20.8)	85.6 (18.7)	.53	1.00 (0.99–1.02)	.52
Mean arterial pressure, mm Hg	86.4 (12.5)	90.2 (13.9)	.03	0.98 (0.96 to <1.00)	.03
Core body temperature, °C	37.0 (36.5, 37.5)	37.3 (36.9, 37.7)	.01	1.01 (0.95–1.18)	.70
Respiratory rate, breaths/ min	16.0 (14.0, 20.0)	15.5 (13.3, 19.5)	.03	1.06 (<1.00 to 1.12)	.05
PEEP, cm H ₂ O	5.0 (5.0, 10.0)	5.0 (5.0, 6.8)	<.01	1.22 (1.07–1.40)	<.01
PaO ₂ /FiO ₂ , mm Hg	277 (161, 378)	350 (263, 412)	<.01	0.99 (0.99 to <1.00)	.03
Urine output, mL/h	100 (50, 193)	100 (56, 175)	.53	1.00 (0.99 to >1.00)	.74
WBC, cells/ μ L	11.7 (9.1, 16.2)	12.5 (9.3, 14.8)	.93	1.03 (0.98–1.08)	.26
Hematocrit, %	29.1 (5.2)	29.7 (5.2)	.40	0.98 (0.93–1.030)	.40
Serum sodium, mmol/L	145 (140, 150)	140 (138, 143)	<.0001	1.04 (1.01–1.08)	.01
Blood urea nitrogen, mg/ dL	20.0 (13.0, 29.0)	20.0 (13.0, 27.5)	.70	1.01 (0.99–1.02)	.35
Serum creatinine, mg/dL	0.70 (0.58, 1.10)	0.62 (0.50, 0.90)	.03	1.31 (<1.00 to 1.85)	.05
Serum bilirubin, mg/dL	0.50 (0.40, 0.80)	0.50 (0.30, 0.80)	.75	0.98 (0.86–1.11)	.72
Maximum serum CPK, U/L	392 (145, 1441)	286 (80, 832)	.10	1.00 (<1.00 to >1.00)	.63
Vasoactive inotropic score* >5	21 (14.0%)	1 (1.2%)	<.001	13.51 (2.75–244.4)	<.001
Receiving sedative/analgesic					
Propofol	42 (28%)	27 (32%)	.51	0.82 (0.46–1.48)	.51
Dexmedetomidine	4 (2.7%)	6 (7.1%)	.17	0.36 (0.09–1.28)	.11
Fentanyl	51 (34%)	26 (31%)	.63	1.15 (0.65–2.06)	.63
Midazolam	33 (22%)	14 (17%)	.33	1.41 (0.72–2.89)	.33
Brain imaging findings					
Subarachnoid hemorrhage	33 (23%)	16 (20%)	0.52	1.25 (0.65–2.49)	.52
Intraparenchymal hemorrhage	44 (31%)	24 (29%)	.72	1.12 (0.62–2.04)	.72
Cerebral infarction	13 (9.2%)	8 (9.8%)	.88	0.93 (0.38–2.45)	.88
Herniation	27 (19%)	22 (27%)	.18	0.65 (0.34–1.24)	.18
Facial trauma	18 (12%)	7 (8.5%)	.38	1.51 (0.62–4.03)	.38

CPK, creatine phosphokinase; DCD, donation after circulatory death; FiO₂, fraction of inspired oxygen; PaO₂, partial pressure of arterial oxygen; PEEP, positive end-expiratory pressure; WBC, white blood cell.

*Vasoactive inotropic score = dopamine ($\mu\text{g} \cdot \text{kg}^{-1} \cdot \text{min}^{-1}$) + dobutamine ($\mu\text{g} \cdot \text{kg}^{-1} \cdot \text{min}^{-1}$) + 100 $\times$ epinephrine ($\mu\text{g} \cdot \text{kg}^{-1} \cdot \text{min}^{-1}$) + 100 $\times$ norepinephrine ($\mu\text{g} \cdot \text{kg}^{-1} \cdot \text{min}^{-1}$) + 10 $\times$ milrinone ($\mu\text{g} \cdot \text{kg}^{-1} \cdot \text{min}^{-1}$) + 10,000 $\times$ vasopressin (units $\cdot \text{kg}^{-1} \cdot \text{min}^{-1}$) + 50 $\times$ levosimendan ($\mu\text{g} \cdot \text{kg}^{-1} \cdot \text{min}^{-1}$).

incomplete data, with multiple imputation used to address missing inputs. In this cohort, the pooled AUC was 0.82 (95% CI 0.78–0.87), indicating maintenance of reasonable discrimination despite missing data. The pooled Brier score was 0.25 (null Brier 0.23), the calibration intercept was -1.71 (95% CI -2.49 to -0.94), and the calibration slope was 1.08 (95% CI 0.56–1.60), indicating systematic overprediction and no improvement over the null model.

Discussion

Model Selection and Performance

We present a novel model, informed by routinely available clinical data, that predicts progression to circulatory death among potential DCD heart donors with an AUC of 0.77 in the development cohort and 0.88 in external validation at a distant US center. In comparison, the DCD III study¹⁷ predicted progression to circulatory death among

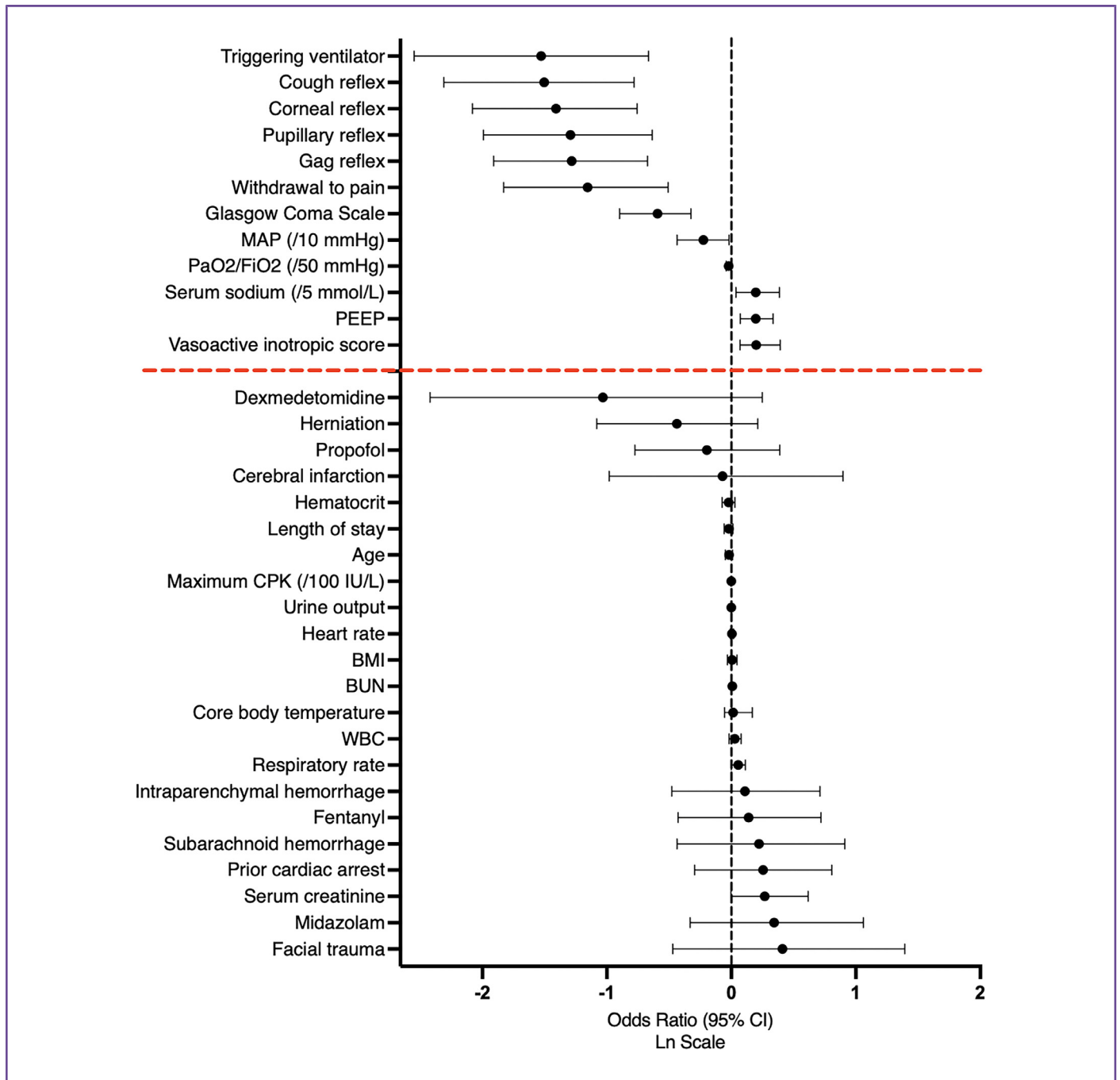


Fig. 2. Odds ratios with associated 95% CIs for the association between pre-withdrawal-of-care clinical variables and a binary outcome of progression to circulatory death among potential DCD heart donors. Variables significantly associated with the outcome of interest are shown above the horizontal red dotted line. BMI, body mass index; BUN, blood urea nitrogen; CI, confidence interval; CPK, creatine phosphokinase; DCD, donation after circulatory death; FiO₂, fraction of inspired oxygen; Ln, natural logarithm; MAP, mean arterial pressure; PaO₂, partial pressure of arterial oxygen; PEEP, positive end-expiratory pressure; WBC, white blood cell.

potential DCD donors of any organ with an AUC of 0.79 to 0.81, a recently published machine-learning model (LightGBM) did so with an AUC of 0.80 to 0.83,²⁴ and the Chinese Donation after Circulatory Death Nomogram²¹ did so with a C statistic of 0.86. Our model thus performs comparably in outcome discrimination to existing models developed for the broader population of potential DCD donors of any organ.

Calibration analysis in external validation demonstrated systematic overprediction of circulatory death, which should be considered when interpreting model predictions. When applied to imputed data, the model maintained good discrimination, although calibration was poor, suggesting model-generated absolute probabilities should be interpreted with caution when using incomplete data.

Table 3 Performance characteristics assessed by 5-fold stratified cross validation for multivariate logistic regression model and 4 machine-learning models predicting progression to circulatory death after withdrawal of care among potential DCD heart donors

Model	Sensitivity	Specificity	AUC	Positive Predictive Value*	Negative Predictive Value*
Multivariate logistic regression	0.827	0.535	0.757	0.761	0.634
Machine learning					
Naïve Bayes	0.597	0.785	0.761	0.830	0.516
Support vector machine	0.800	0.464	0.742	0.727	0.565
Random forest	0.800	0.427	0.713	0.714	0.545
Decision tree	0.687	0.570	0.628	0.740	0.505

Performance metrics shown are the average of 5 validation replicates.

AUC, area under the receiver operating characteristic curve; DCD, donation after circulatory death.

*Predictive values calculated assuming the outcome prevalence observed in the model development cohort.

Although multiple models have been developed to predict progression to circulatory death among DCD donors broadly, to our knowledge this study represents the first such model derived exclusively from a cohort of potential heart donors. Fewer than 5% of all DCD donors donate a heart, underscoring the more stringent physiologic and logistical requirements for heart donation and the unique nature of this subgroup of donors.²⁵ For example, the mean donor age in our development and validation cohorts was 37 and 34 years, respectively, compared with a mean donor age greater than 55 years in the DCD III study and a median donor age of 54 in the LightGBM model-derivation cohort. Beyond more stringent donor-age preferences, eligibility for heart donation requires preserved cardiovascular function prior to withdrawal of life-sustaining therapy and imposes narrower tolerances for functional warm ischemic time given the heart's limited ischemic tolerance relative to other transplantable organs.¹¹ Together, these factors suggest that models derived from heterogeneous DCD cohorts, predominantly composed of nonheart organ donors, may not be generalizable to the population of potential heart donors.

As a result of organ scarcity, transplant teams may be reluctant to decline to pursue heart procurement even if the chance of donor progression to circulatory death is low.¹⁷ In this context, it may be appropriate to optimize for negative predictive value at the expense of overall model accuracy, seeking to reliably identify the subset of offers that are exceedingly unlikely to progress to circulatory death and for which procurement can justifiably be declined. With this in mind, the 100% negative predictive value achieved by our model in external validation represents a significant strength and a promising potential avenue to reduce resource expenditure on unsuccessful procurement without negatively affecting organ use rates. Although we selected a probability threshold that balanced model accuracy with a desire to limit false-negative predictions, we report model sensitivity and specificity across a range of thresholds to illustrate the relevant tradeoffs in clinical application.

Although model performance was robust in external validation, discrimination remained imperfect, suggesting

the presence of unaccounted-for factors influencing circulatory death. For example, the significant influence exerted on the outcome of interest by the clinician responsible for withdrawal of care, namely via the administration of palliative sedative and analgesic medications after extubation, cannot be accounted for in this study and generally complicates attempts to predict successful DCD donation. Standardization of withdrawal-of-care practices across centers would likely improve the predictability of the heart procurement process.

Among candidate model types, multivariate logistic regression emerged as the most suitable for our purpose, given a greater negative predictive value (63.4%) compared with naïve Bayes (51.6%), as well as the greater interpretability of logistic regression compared with machine-learning models. However, the optimal modeling technique for classification tasks depends on the size and complexity of the development dataset,²⁷⁻²⁹ and with larger datasets a machine-learning algorithm may prove more effective.

Variables Influencing Circulatory Death

We found that variables associated with progression to circulatory death were generally those indicative of donor brainstem function, respiratory support requirements, and hemodynamic stability. Many, although not all, variables previously associated with donor progression to circulatory death appear to be conserved between heart donors and donors at large. Similar to previous studies of potential DCD donors broadly, we found brainstem reflexes,^{15-17,19,23,24} GCS,²⁴ mean arterial pressure,²⁴ vasoactive medication requirement,^{16-18,20} spontaneous respiration,^{17,18,20,22-24} respiratory support requirement,^{15,19,21,24} and serum sodium²⁴ to be associated with progression to circulatory death.

In contrast, we did not find donor age^{18,22} to be associated with circulatory death, perhaps because of the younger average age of heart donors compared with donors of other organs. We also did not find body mass index,^{23,24} history of cardiac arrest,²⁴ sedative/analgesic use,¹⁵ or

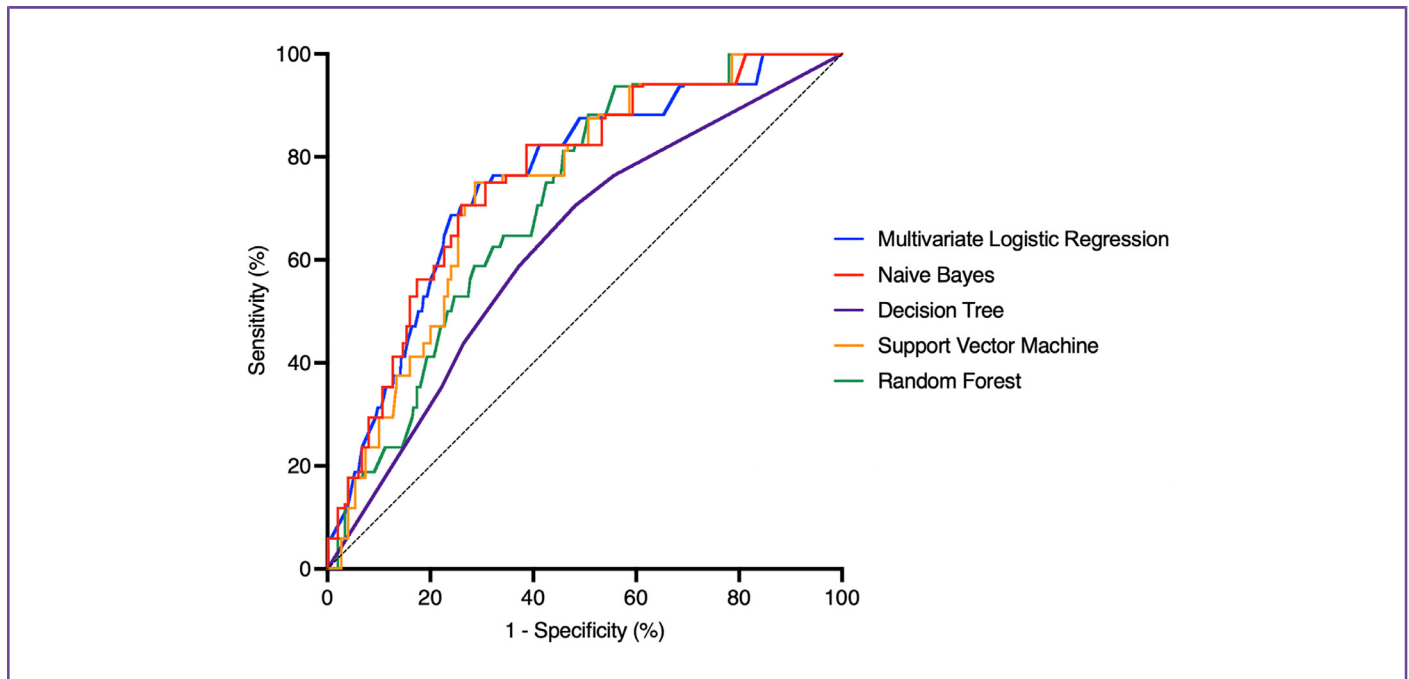


Fig. 3. Average receiver operating characteristic curves across 5-fold stratified cross-validation for candidate model types developed to predict progression to circulatory death after withdrawal of care among potential donation after circulatory death heart donors.

brain imaging findings^{17,21} to be informative; whether these discrepancies reflect intrinsic differences between heart donors and nonheart donors, or simply differences in study cohorts or methodology, is not clear. Hospital length of stay^{17,21} and respiratory rate¹⁸ were significantly associated with progression to circulatory death by descriptive analysis, although not by univariate logistic regression.

Certain variables previously associated with progression to circulatory death, such as mechanism of

intubation¹⁸ and findings of swirl sign or cisterna ambiens on brain computed tomography,²¹ could not be evaluated in our study because of the lack of relevant data recording in UNOS. Other previously described variables of influence, such as cardiovascular comorbidity,²⁰ are less relevant to the subset of potential heart donors.

In univariate analysis, VIS was significantly associated with progression to circulatory death both as a continuous variable and as a binary cutoff of VIS >5. Among different

Table 4 Component variables with associated model coefficients for multivariate logistic regression model predicting progression to circulatory death after withdrawal of care among potential DCD heart donors; model sensitivity and specificity across a range of probability thresholds

Variable	B	SE	Wald	Significance	Odds (95% CI)
Constant	2.12	0.78	—	—	—
Vasoactive inotropic score >5	2.45	1.16	4.45	.04	11.53 (1.19–111.78)
Triggering ventilator	−0.87	0.47	3.42	.07	0.42 (0.17–1.05)
Presence of pupillary reflex	−0.80	0.35	5.21	.02	0.45 (0.23–0.89)
GCS (range 3–15)	−0.37	0.12	10.46	<.01	0.69 (0.55–0.86)
PEEP, cm H ₂ O	0.18	0.07	6.35	.01	1.20 (1.04–1.38)
Probability threshold	Sensitivity		Specificity		Youden's Index
0.2	1.00		0.05		0.05
0.3	0.95		0.18		0.13
0.4	0.91		0.31		0.22
0.5	0.85		0.52		0.38
0.6	0.71		0.70		0.42
0.7	0.60		0.83		0.43
0.8	0.44		0.90		0.34

CI, confidence interval; DCD, donation after circulatory death; GCS, Glasgow Coma Scale; PEEP, positive end-expiratory pressure; SE, standard error; VIS, vasoactive inotropic score.

Youden's index = sensitivity + specificity − 1.

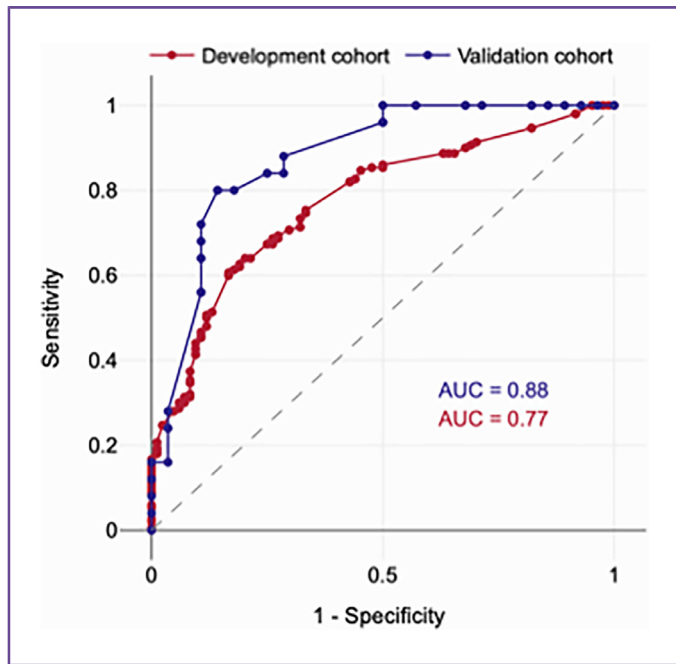


Fig. 4. Receiver operating characteristic curves for development and validation of a multivariate logistic regression model predicting progression to circulatory death after withdrawal of care among potential donation after circulatory death heart donors. AUC, area under the receiver operating characteristic curve.

integer cutoffs, VIS >5 achieved the best discrimination and was selected as the candidate model feature. In multivariate logistic regression, a model incorporating the binary variable VIS >5 outperformed a model incorporating VIS as a continuous variable (AUC 0.772 vs 0.770; sensitivity 52.4% vs 50.0%), and thus VIS >5 was retained as the model feature.

Limitations

This study is limited by the delimited, incomplete, and nonstandardized reporting of donor data in UNOS, which may impair identification of associations between relevant variables and progression to circulatory death. The lack of consistent documentation of the time elapsed before procurement abandonment limits our ability to predict circulatory death within discrete time windows, as has been done in other studies. Because of the retrospective nature of the study, data collection and clinical protocols were not standardized, which may introduce unknown biases.

Multiple model inputs are measurements of donor brainstem function and thus likely exhibit some degree of collinearity, introducing the possibility of unstable coefficient estimates in model development. The confidence interval for the regression coefficient of the VIS model input is very wide, suggesting that the estimated effect size may be unstable and sensitive to the study sample. Although both the model development and validation cohorts included offers from various donor hospitals spanning multiple organ procurement organizations, it is possible that withdrawal-of-care practice differs meaningfully across regions,³⁰ and model performance in distant centers may vary.

In external validation, the exclusion of offers with incomplete data may have skewed the cohort composition and may account for the discrepant rate of progression to circulatory death between the development and validation cohorts (64.1% vs 47.2%). In addition, the lack of imputed data in the validation cohort may account for the observed improvement in AUC between development and validation, which is atypical in model development.

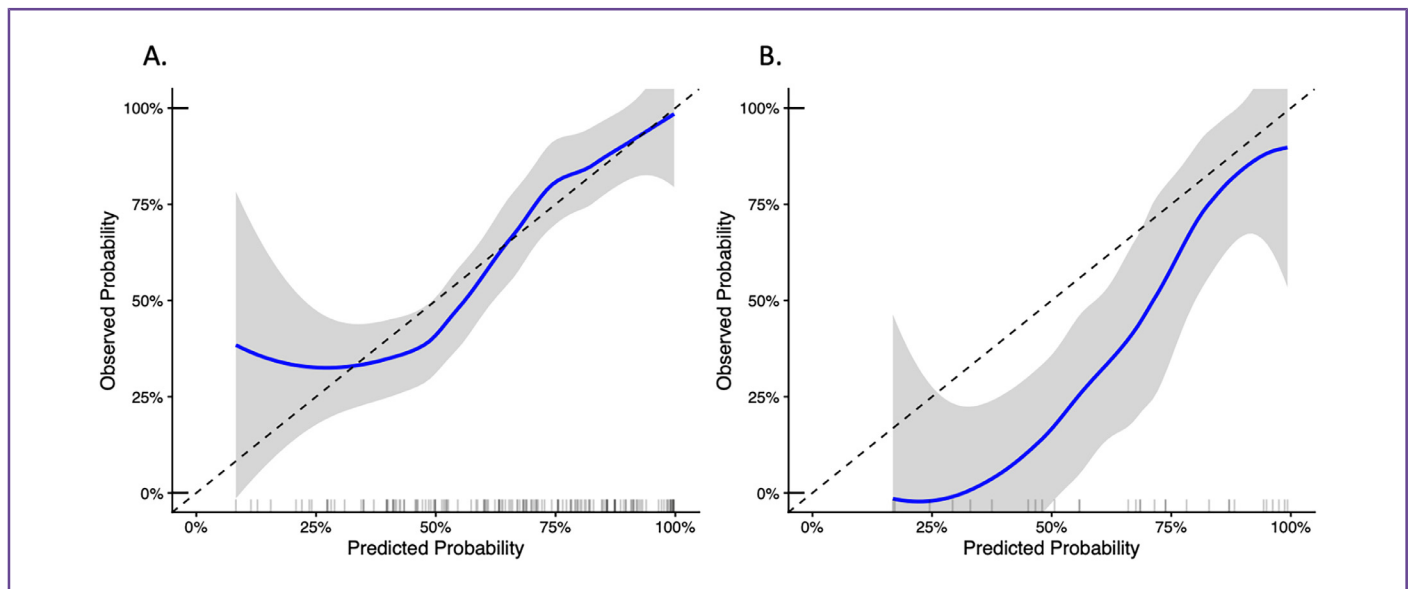
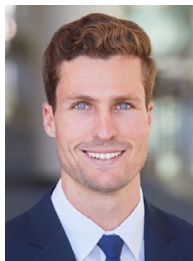


Fig. 5. Calibration plots for a novel multivariate logistic regression model predicting progression to circulatory death after withdrawal of care among potential donation after circulatory death heart donors in (A) development cohort and (B) external validation cohort at a distant US center. Associated rug plots displaying the distribution of model-predicted probabilities are shown on the x axes.

Finally, as discussed previously, nonstandardized withdrawal-of-care protocols across donor centers and the lack of documentation of palliative medication administration during withdrawal of care limit the ability to predict donor progression to circulatory death using data currently available at the time of organ offer.

Conclusions

We present a novel multivariate logistic regression model, developed using readily available clinical data and validated at a distant center, that predicts progression to circulatory death among potential DCD heart donors with accuracy comparable to published models developed for DCD donors broadly. To our knowledge, this is the first reported model developed exclusively in potential DCD heart donors, and may inform efforts to improve resource allocation in DCD heart procurement. An online version of our model is available for experimental use at <https://dcdheartprediction.com/>



Austin Ayer

Disclosures

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CRedit authorship contribution statement

AUSTIN AYER: Writing – review & editing, Writing – original draft, Visualization, Project administration, Methodology, Data curation, Conceptualization. **PRANEET MYLAVARAPU:** Formal analysis, Data curation, Conceptualization. **DAVID GOLOMBECK:** Data curation. **BIJAL RAJPUT:** Data curation. **FADI AZAR:** Methodology, Data curation. **BERNARD S. KADOSH:** Methodology, Data curation. **QUAN M. BUI:** Writing – review & editing, Methodology. **MARK J. KEARNS:** Writing – review & editing, Conceptualization. **VICTOR PRETORIUS:** Writing – review & editing, Methodology, Conceptualization.

MARCUS A. UREY: Writing – review & editing, Methodology, Conceptualization. **ANTOINETTE S. BIRS:** Writing – review & editing, Project administration, Methodology, Data curation, Conceptualization. **ERIC D. ADLER:** Writing – review & editing, Supervision, Methodology, Conceptualization.

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Patient consent

This study was conducted in accordance with the ethical principles of the Declaration of Helsinki as revised in 2024. The study protocol was approved by the Institutional Review Board of the University of California, San Diego (IRB# 810593, determination May 13, 2024) and was determined to be not human subject's research. New York University contributed a de-identified dataset under a data use agreement; the New York University Institutional Review Board determined that its contribution did not constitute human subjects' research.

Supplementary materials

Supplementary material associated with this article can be found in the online version at [doi:10.1016/j.cardfail.2026.05.017](https://doi.org/10.1016/j.cardfail.2026.05.017).

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